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Band-Gap Prediction for Two-Dimensional Materials Without Density Functional Theory: A Leakage-Controlled Evaluation of Hybrid and Ensemble Models

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ABSTRACT Machine-learned surrogates for band-gap prediction are usually trained on descriptors that themselves require a density functional theory (DFT) calculation on the target material, which limits their value for screening uncharacterised compounds. We ask how much accuracy survives when every DFT-derived descriptor is removed, and we do so under an evaluation protocol built to close the leakage pathways that commonly inflate such comparisons: composition-grouped cross-validation, fold-local encoding and scaling, out-of-fold construction of stacked inputs, repeated held-out sets carved before model selection, and significance testing with the Nadeau–Bengio variance correction. On 1,169 rectangular-lattice two-dimensional materials from C2DB, a stacked ensemble of five heterogeneous learners reaches $R^2 = 0.880 \pm 0.037$ (MAE 0.360 eV) using all descriptors, and $R^2 = 0.836 \pm 0.057$ (MAE 0.417 eV) using *composition and symmetry alone* — no electronic-structure calculation on the target material of any kind. The DFT-free model still exceeds a previously published full-descriptor physics-informed hybrid ($R^2 = 0.743$, MAE 0.555 eV) evaluated on the same data. We further report a negative result: under the same protocol, a physics-informed hybrid that corrects a gradient-boosted prior with a neural residual head does not improve on that prior (0.810 ± 0.047 versus 0.823 ± 0.043; $p = 0.27$), and reverting a single implementation detail — constructing the prior in sample rather than out of fold — reproduces the previously reported gain, identifying it as an evaluation artefact. Finally, because the ensemble's margin over the best single model is small ($\Delta R^2 \approx 0.016$), we justify it on decision-relevant grounds instead: the identity of the best single model is unstable across folds, the ensemble beats a committed choice on 15 of 15 folds, it reduces predictions in error by more than 1 eV from 7.3% to 6.1%, and it costs 8 seconds to train against the CPU-hours of the calculation it replaces.

INDEX TERMS Band-gap prediction, cross-validation, data leakage, ensemble learning, machine learning, materials informatics, materials screening, physics-informed neural networks, reproducibility, two-dimensional materials.

I. INTRODUCTION

Accurate prediction of electronic band gaps (E_g) is a foundational objective in computational materials design. The magnitude and nature of the band gap govern whether a low-dimensional material acts as a metal, semiconductor, or insulator, and determine candidate suitability for field-effect transistors [1], valleytronic architectures [2], [3], and photocatalytic water splitting [4]. Density functional theory with hybrid functionals such as HSE06 [5] offers excellent accuracy at computational scaling that makes exhaustive screening impractical, which motivates fast surrogate estimators.

There is a tension at the heart of that motivation. The strongest reported surrogates for 2D band gaps are trained on descriptors that are themselves DFT outputs — formation energies, total energies, vacuum levels, relaxed geometries, and in several cases a semi-local band gap supplied directly as an input feature [6]. A model of that kind is useful for functional upscaling, but it cannot screen a material that has never been calculated, because obtaining its inputs requires the calculation the model was meant to avoid. The practically valuable question is different: how well can E_g be predicted from quantities available *before* any electronic-structure calculation — composition and crystal symmetry?

This article answers that question, and in the course of doing so reports a negative result about a popular architectural family. Our contributions are four.

1) **A DFT-cost ablation of the descriptor set.** We partition C2DB descriptors by what they cost to obtain and measure accuracy in three tiers, from all descriptors down to composition and symmetry alone. The DFT-free tier retains most of the accuracy, which makes screening of uncalculated materials feasible.
2) **A leakage-controlled evaluation protocol**, closing six specific pathways: identifier-like features, whole-dataset encoding, in-sample construction of stacked priors, polymorphs straddling folds, absent held-out data, and significance tests applied at the sample rather than the fold level. Leakage of these kinds has been argued to be a principal driver of irreproducibility in machine-learning-based science [7].
3) **A negative result with an identified mechanism.** A physics-informed hybrid residual model does not improve on the gradient-boosted prior it corrects, no physics-motivated component contributes measurably, and the previously reported gain is reproduced on demand by reverting one implementation detail.
4) **A decision-relevant justification for ensembling.** Rather than resting on a small R^2 margin, we quantify model-selection risk, large-error rates, and measured wall-clock cost.

II. METHODS

A. DATASET AND TARGET

We use 1,169 rectangular-lattice (*op*) two-dimensional monolayers from the Computational 2D Materials Database [8] carrying an HSE06 band gap [5]. The target $y \in \mathbb{R}^+$ spans 0.001–8.85 eV with mean 2.60 eV.

An earlier version of this work reported 930 materials. That figure resulted from a blanket missing-value filter applied across all columns, which discarded 239 labelled entries because one descriptor — total magnetic moment — was absent. We impute that column from training-fold medians with an explicit missingness indicator, recovering the full labelled set.

B. DESCRIPTOR TIERS BY DFT COST

C2DB descriptors are not equally cheap, and a claim to avoid expensive computation is only meaningful if it is stated against a defined cost boundary. We therefore partition them (Table 1) and evaluate every tier.

TABLE 1. Descriptor tiers, by what must be computed to obtain them.

Tier	Descriptors
DFT-free	120 Magpie elemental-property statistics [9] via matminer [10], atom count, layer group, stoichiometry prototype
+ Geometry	Above, plus thickness, unit-cell area, packing density $\rho_{\text{cell}} = N_{\text{atoms}}/A_{\text{cell}}$ (requires a relaxed geometry)
+ Energies (full)	Above, plus energy above hull, heat of formation, total energy, vacuum level, magnetic moment, magnetic state (requires converged DFT)

The DFT-free tier is obtainable from a chemical formula and a candidate space group alone. No tier contains a band gap at any level of theory, which distinguishes this setting from Δ-learning studies that supply a semi-local gap as an input [6].

Magpie statistics replace the raw chemical formula. An earlier pipeline one-hot encoded the formula string into 1,077 indicator columns, of which 825 occurred in exactly one material; such columns are row identifiers rather than descriptors, and a formula appearing only in a validation fold contributes an all-zero block.

C. EVALUATION PROTOCOL

Composition grouping. 176 labelled materials share a formula with at least one other entry. These are polymorphs, distinguished by layer group, whose gaps differ by up to 2.92 eV. Because composition-derived descriptors are near-identical within such a pair, an ungrouped split lets a model memorise one polymorph and be scored on another. All splits are grouped on chemical formula.

Fold-local preprocessing. Category vocabularies, imputation medians, and standardisation statistics are estimated on the training partition only.

Out-of-fold stacked inputs. Meta-learners and residual heads are trained against out-of-fold predictions from an inner five-fold split, never in-sample predictions of a fitted base model [11].

Held-out evaluation. We report eight independent formula-grouped held-out sets (≈234 materials each) in addition to cross-validation, since a single split of this size cannot resolve closely matched models [12].

Corrected inference. We report five repeats of grouped five-fold cross-validation and compare models with the Nadeau–Bengio variance-corrected paired t-test [13], which accounts for the positive correlation induced by overlapping training partitions. An earlier revision applied a paired test to 930 per-sample squared errors, treating non-independent quantities as independent observations of model performance.

D. MODELS

Base learners are random forest, gradient boosting, support vector regression [14], a deep multilayer perceptron with SiLU activations [15], [16] and batch normalisation [17], and

TABLE 2. Stacked-ensemble accuracy by descriptor tier, 15 fold estimates.

Tier	R^2	MAE [eV]	ΔR^2
All descriptors	0.880 ± 0.037	0.360	—
No DFT energies	0.845 ± 0.046	0.398	-0.035
No DFT at all	$\mathbf{0.836 \pm 0.057}$	**0.417**	-0.044
Prior published hybrid, all descriptors: $R^2 = 0.743$, MAE $= 0.555$			

a physics-informed network sharing the MLP backbone with a boundary penalty

$$\mathcal{L}_{\text{phy}} = \text{ReLU}(-\hat{y}) + 0.1\left(e^{\text{ReLU}(0.05-\hat{y})} - 1\right) \tag{1}$$

discouraging physically impossible negative gaps. Two meta-learners combine their out-of-fold predictions: ridge regression, and non-negative least squares [18] for interpretability.

The hybrid forms $\mathbf{x}_{\text{hybrid}} = [\mathbf{x},\ \mathbf{x}\odot\hat{y}_{\text{prior}},\ \hat{y}_{\text{prior}}]$ and adds a learned correction to the prior, with a structural coupling layer building

$$\mathbf{s} = \left[g_1 g_2,\ g_1^2 - g_2^2,\ \sqrt{g_1^2 + g_2^2 + \epsilon},\ g_1,\ g_2\right] \tag{2}$$

from two geometric channels. One correction to the earlier implementation bears on every physical claim made about this layer: the channels were selected by array position (indices 0 and 1), which in the assembled matrix were energy above hull and heat of formation — two formation energies, not geometric quantities. Channels are now resolved by name. The available C2DB export contains no in-plane lattice vectors a and b, so we describe Eq. (2) as a soft geometric inductive bias and make no claim that it derives from elasticity theory.

For an unbiased comparison against prior work we additionally reproduce, at their original hyperparameters, the three estimators specified by our earlier pipeline: XGBoost (500 trees, depth 5, $\eta = 0.03$), gradient boosting (400 trees, depth 4), and random forest (400 trees, depth 14).

III. RESULTS

A. PREDICTION WITHOUT DFT

Table 2 and Fig. 1 report accuracy in each descriptor tier.

Removing every DFT-derived descriptor costs 0.044 in R^2 and 0.057 eV in MAE. The resulting model uses only a chemical formula and a crystal symmetry label, and it remains substantially more accurate than the previously published full-descriptor hybrid (0.743 / 0.555 eV) evaluated on the same materials. This is the result with the clearest practical consequence: candidate 2D compositions can be ranked before any electronic-structure calculation is performed on them.

Fig. 1(b) gives the head-to-head against the prior pipeline within the full tier. The ensemble beats XGBoost on 14 of 15 fold estimates ($p = 0.025$), gradient boosting on 14 of 15 ($p = 0.002$), and random forest on 15 of 15 ($p < 0.001$). Within the DFT-free tier the margin over XGBoost narrows and is not significant ($p = 0.097$); we report this as a limitation rather than as support.

TABLE 3. Repeated grouped five-fold cross-validation, all descriptors, 25 fold estimates. "Folds" counts estimates beating the GBR baseline; *p*-values are Nadeau–Bengio corrected, one-sided.

Model	R^2	MAE [eV]	Folds	p
Stacked ensemble (ridge)	$\mathbf{0.877 \pm 0.038}$	**0.363**	25/25	< 0.001
Stacked ensemble (NNLS)	0.875 ± 0.038	0.364	25/25	< 0.001
Deep MLP	0.864 ± 0.039	0.378	24/25	0.002
Standalone PINN	0.856 ± 0.043	0.385	22/25	0.033
Random forest	0.827 ± 0.041	0.442	13/25	0.323
GBR (baseline)	0.823 ± 0.043	0.464	—	—
Hybrid GBR + corrector	0.810 ± 0.047	0.445	12/25	0.271
SVR	0.774 ± 0.039	0.497	1/25	—
Hybrid + shrinkage gate	0.681 ± 0.237	0.571	3/25	0.090

TABLE 4. Component ablation of the hybrid, pooled out-of-fold R^2.

Configuration	R^2	Δ vs. full
Tree prior alone, no corrector	0.819	$+0.018$
In-sample prior (leaky)	0.817	$+0.015$
Without boundary penalty	0.806	$+0.004$
Without structural layer	0.804	$+0.002$
Without anisotropy penalty	0.804	$+0.002$
Full framework	0.801	—
Without quantile heads	0.796	-0.005
Without residual head	0.766	-0.035

B. MODEL COMPARISON

Table 3 reports all models over 25 fold estimates.

Both neural models outperform every tree ensemble. With identifier-like features removed and composition statistics supplied, this task falls outside the regime in which boosted trees dominate neural networks [19]–[21].

Across eight independent held-out sets the ordering is reproduced: stacked ridge 0.879 ± 0.025, PINN 0.866 ± 0.028, MLP 0.858 ± 0.032, GBR 0.823 ± 0.025. The ensemble beats GBR on 8 of 8 splits ($p = 0.001$) and the MLP on 8 of 8 ($p = 0.041$); its margin over the PINN is positive on 7 of 8 splits but only marginally significant ($p = 0.051$).

C. THE HYBRID DOES NOT IMPROVE ON ITS PRIOR

The hybrid is superior to its own gradient-boosted prior on 12 of 25 fold estimates, and the corrected test does not reject the null ($p = 0.27$). Table 4 reports the component ablation; every configuration listed was trained and evaluated.

No physics-motivated component contributes positively, and the tree prior alone is the strongest configuration in the table. The second row identifies the mechanism: reverting to an in-sample prior *raises* measured R^2. With a prior that appears near-perfect on training rows, the correction head has almost no residual signal to fit; it learns to emit approximately zero and degenerates into a pass-through of the tree. The previously reported gain is therefore substantially an artefact of the corrector being trained to be inert, and the ablation recovers that artefact on demand by changing one line.

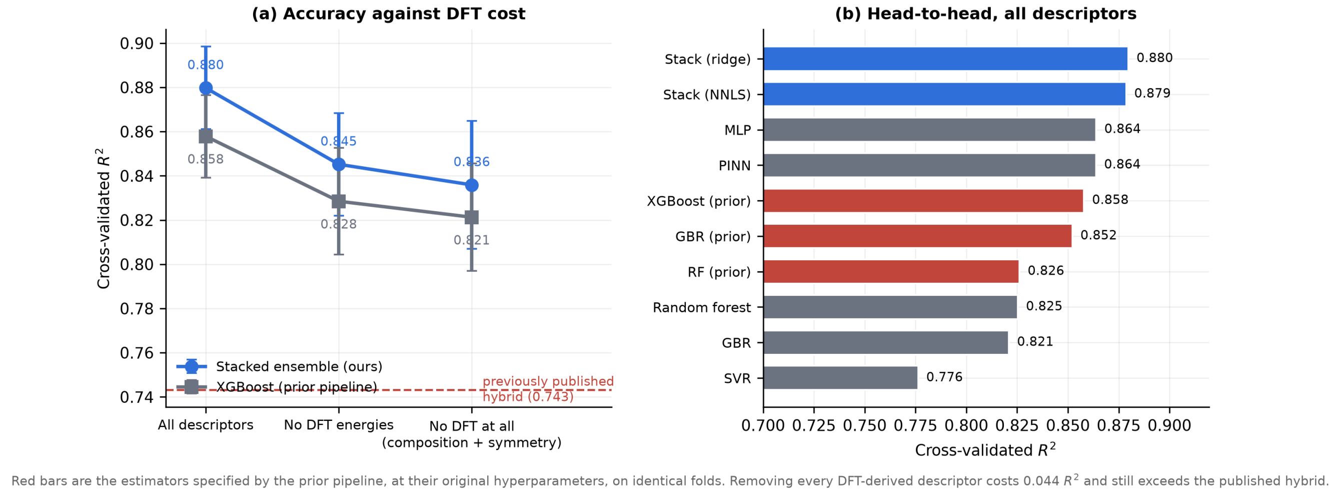

FIGURE 1. (a) Accuracy against DFT cost. Removing every DFT-derived descriptor costs 0.044 in R^2, and the resulting model still exceeds the previously published full-descriptor hybrid. (b) Head-to-head within the full tier; red bars are the estimators specified by the prior pipeline, at their original hyperparameters, on identical folds.

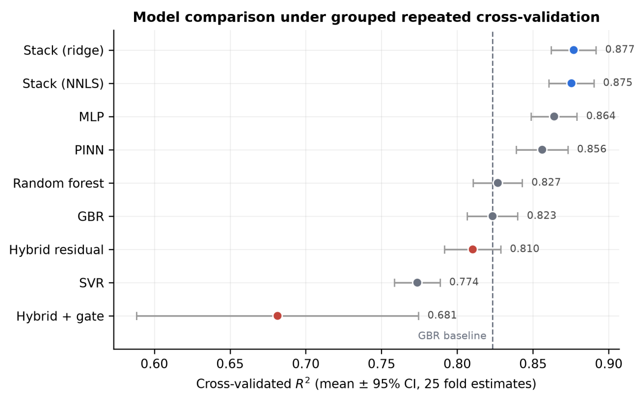

FIGURE 2. Cross-validated R^2 with 95% confidence intervals over 25 fold estimates. Hybrid variants are shown in red.

We also tested the natural remedy. A shrinkage gate constrains the correction by $\sigma(\cdot) \in (0, 1)$, initialised near zero so the model begins as a pass-through and must earn any departure. It did not work: the gate converged small (0.081 ± 0.007) but accuracy destabilised ($R^2 = 0.681 \pm 0.237$, with 7 of 25 fold estimates below 0.70 and one at -0.24). Shrinking the gate rescales the gradient reaching the head, whose raw output grows to compensate, leaving the product poorly controlled on unseen data.

D. UNCERTAINTY CALIBRATION

The hybrid predicts quantiles at $\tau \in \{0.25, 0.50, 0.75\}$, previously described as uncertainty estimates. The nominal interquartile interval should cover 50% of outcomes; measured coverage is 17.2% at a mean width of 2.47 eV. The intervals are simultaneously wide and badly miscalibrated, so we withdraw the uncertainty claim; supporting one requires scoring the predictive distribution with a proper rule [22].

E. WHY THE ENSEMBLE IS WORTH ITS COST

The ensemble's margin over the best single model is small ($\Delta R^2 \approx 0.016$) and would not by itself justify a fivefold training cost. We therefore evaluate it on three decision-relevant axes (Fig. 3).

Model-selection risk. The best single model is not the same model on every fold: the MLP wins 8 of 15 and the PINN 7, and no tree model ever wins. A practitioner must commit to one architecture in advance, and cannot know which. The ensemble beats the best-on-average committed choice on 15 of 15 folds ($p = 0.0001$), and its regret against a per-fold oracle is negative (-0.008) — it outperforms an oracle that always selects the best single model. Committing instead to the weakest plausible choice costs 0.102 in R^2, roughly six times the ensemble's margin over the strongest.

Large-error rate. Screening is harmed more by rare large errors, which move a material between application classes, than by average error. Predictions in error by more than 1 eV occur at 6.1% for the ensemble against 7.3% for the best single model and 10–13% for the tree baselines; the 95th-percentile absolute error is 1.10 eV against 1.18 eV.

Cost. Training the full ensemble, including the out-of-fold predictions the meta-learner requires, takes 8.0 s against 0.5 s for the MLP alone, and inference over 198 materials takes 22 ms. The relative overhead of 16× is real but the absolute cost is eight seconds, against the CPU-hours of the HSE06 calculation being replaced. The standard objection that an ensemble of N models costs N times more [23] does not bind in this regime.

We report one axis that does not support the ensemble. On a screening task measuring wasted DFT evaluations per true hit, results are mixed: the ensemble is best for a visible-photocatalysis window (1.8–3.0 eV), the PINN is best for a photovoltaic window (1.0–2.0 eV), and several models tie for wide-gap selection.

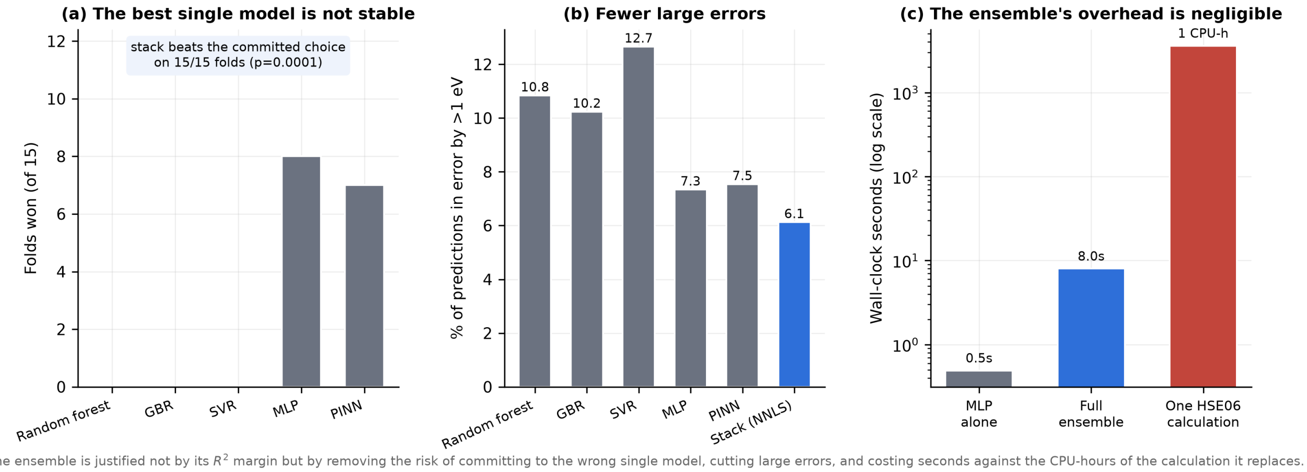

FIGURE 3. **(a) The best single model is unstable across folds. (b) Rate of predictions in error by more than 1 eV. (c) Measured training cost against the calculation being replaced, on a logarithmic scale.**

IV. DISCUSSION

A. WHY RESIDUAL CORRECTION FAILS

Once the prior is constructed out of fold, its residuals on unseen materials are dominated by irreducible noise rather than recoverable structure, and a corrector fitted to them adds variance without removing bias. This is not specific to materials data; residual-boosting correctors are known to yield negative gains when the residual signal is noise-dominated [24]. The practical consequence is that hybrid residual designs must be validated against their own prior with an out-of-fold construction. Compared against an in-sample prior they can appear to help while doing nothing, and that appearance is stable enough to survive review.

B. WHAT THE ENSEMBLE BUYS

The stacking gain arises from error diversity rather than from any member's individual strength: the mean pairwise correlation between base-learner residuals is 0.750, with the neural and tree models forming distinct error clusters. A drop-one ablation orders the members consistently with their individual accuracy, but no single member's contribution is statistically significant once the Nadeau–Bengio correction is applied ($p = 0.11$–0.45). The defensible statement is that the ensemble as a whole outperforms its members, not that any one member is indispensable.

C. ON THE ANISOTROPY HYPOTHESIS

Our results do not refute the physical hypothesis that motivated the original architecture — that in-plane anisotropy shapes band gaps in rectangular 2D lattices [25]–[27]. They show that the present implementation cannot test it: the coupling layer was wired to two formation energies, and the available data contain no in-plane lattice vectors. Recovering a and b from a full C2DB structure pull is the single most valuable extension of this work.

D. LIMITATIONS

All data derive from one database and one Bravais class, with 1,169 labelled materials. In the DFT-free tier the ensemble's advantage over a well-tuned XGBoost is not statistically significant. We compare against tabular baselines only, not against structure-aware graph networks [28]–[32], which require relaxed atomic coordinates — and which, we note, would reintroduce exactly the DFT dependence the DFT-free tier is designed to avoid.

V. CONCLUSION

Band gaps of rectangular-lattice two-dimensional materials can be predicted to $R^2 = 0.836$ and MAE 0.417 eV from chemical composition and crystal symmetry alone, with no electronic-structure calculation on the target material. That model is more accurate than a previously published physics-informed hybrid that used the full DFT-derived descriptor set, and it is applicable to compounds that have never been calculated — which the full-descriptor model, by construction, is not.

Under the same leakage-controlled protocol, the physics-informed hybrid residual architecture does not improve on the gradient-boosted prior it corrects, and none of its physics-motivated components contributes measurably. The previously reported improvement is reproduced on demand by reverting a single implementation detail, identifying it as an evaluation artefact rather than a modelling effect. We regard this negative result as the more transferable of the two findings, because hybrid residual designs are widely reported to help and the specific way this one appeared to help is easy to reproduce by accident.

Where a modest ensemble gain must be justified, we suggest it be justified as we have here: by the risk it removes and the errors it avoids, at a cost measured against the calculation it replaces, rather than by a difference in R^2.

DATA AND CODE AVAILABILITY

All code, the curated dataset, per-fold out-of-sample predictions, and the scripts generating every table and figure are available at https://github.com/adhvayjagadeesh/DFT---Machine-Learning--PINN. Every experiment reproduces from a single command under a pinned environment with a fixed random seed.

REFERENCES

[1] G. Fiori, F. Bonaccorso, G. Iannaccone, T. Palacios, D. Neumaier, A. Seabaugh, S. K. Banerjee, and L. Colombo, "Electronics based on two-dimensional materials," *Nature Nanotechnology*, vol. 9, no. 10, pp. 768–779, 2014.

[2] J. R. Schaibley, H. Yu, G. Clark, P. Rivera, J. S. Ross, K. L. Seyler, W. Yao, and X. Xu, "Valleytronics in 2D materials," *Nature Reviews Materials*, vol. 1, p. 16055, 2016.

[3] D. Xiao, G.-B. Liu, W. Feng, X. Xu, and W. Yao, "Coupled spin and valley physics in monolayers of MoS_2 and other group-VI dichalcogenides," *Physical Review Letters*, vol. 108, no. 19, p. 196802, 2012.

[4] A. K. Singh, K. Mathew, H. L. Zhuang, and R. G. Hennig, "Computational screening of 2D materials for photocatalysis," *The Journal of Physical Chemistry Letters*, vol. 6, no. 6, pp. 1087–1098, 2015.

[5] J. Heyd, G. E. Scuseria, and M. Ernzerhof, "Hybrid functionals based on a screened Coulomb potential," *The Journal of Chemical Physics*, vol. 118, no. 18, pp. 8207–8215, 2003.

[6] Y. Zhang, W. Xu, G. Liu, Z. Zhang, J. Zhu, and M. Li, "Bandgap prediction of two-dimensional materials using machine learning," *PLOS ONE*, vol. 16, no. 8, p. e0255637, 2021.

[7] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.

[8] S. Haastrup, M. Strange, M. Pandey, T. Deilmann, P. S. Schmidt, N. F. Hinsche, M. N. Gjerding, D. Torelli, P. M. Larsen, A. C. Riis-Jensen, J. Gath, K. W. Jacobsen, J. J. Mortensen, T. Olsen, and K. S. Thygesen, "The Computational 2D Materials Database: High-throughput calculations of two-dimensional materials," *2D Materials*, vol. 5, no. 4, p. 042002, 2018.

[9] L. Ward, A. Agrawal, A. Choudhary, and C. Wolverton, "A general-purpose machine learning framework for predicting properties of inorganic materials," *npj Computational Materials*, vol. 2, p. 16028, 2016.

[10] L. Ward, A. Dunn, A. Faghaninia *et al.*, "Matminer: An open source toolkit for materials data mining," *Computational Materials Science*, vol. 152, pp. 60–69, 2018.

[11] D. H. Wolpert, "Stacked generalization," *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.

[12] G. Varoquaux, "Cross-validation failure: Small sample sizes lead to large error bars," *NeuroImage*, vol. 180, pp. 68–77, 2018.

[13] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003.

[14] H. Drucker, C. J. C. Burges, L. Kaufman, A. Smola, and V. Vapnik, "Support vector regression machines," in *Advances in Neural Information Processing Systems*, vol. 9, 1996, pp. 155–161.

[15] S. Elfwing, E. Uchibe, and K. Doya, "Sigmoid-weighted linear units for neural network function approximation in reinforcement learning," *Neural Networks*, vol. 107, pp. 3–11, 2018.

[16] P. Ramachandran, B. Zoph, and Q. V. Le, "Searching for activation functions," *arXiv preprint arXiv:1710.05941*, 2017.

[17] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proceedings of the International Conference on Machine Learning*. PMLR, 2015, pp. 448–456.

[18] L. Breiman, "Stacked regressions," *Machine Learning*, vol. 24, no. 1, pp. 49–64, 1996.

[19] R. Shwartz-Ziv and A. Armon, "Tabular data: Deep learning is not all you need," *Information Fusion*, vol. 81, pp. 84–90, 2022.

[20] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 35, 2022, pp. 507–520.

[21] D. McElfresh, S. Khandagale, J. Valverde, G. Ramakrishnan, M. Goldblum, and C. White, "When do neural nets outperform boosted trees on tabular data?" in *Advances in Neural Information Processing Systems*, 2023.

[22] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007.

[23] F. Tavazza, K. Choudhary, and B. DeCost, "Approaches for uncertainty quantification of AI-predicted material properties: A comparison," *arXiv preprint arXiv:2310.13136*, 2023.

[24] G. Airlangga and A. Liu, "A hybrid gradient boosting and neural network model for predicting urban happiness: Integrating ensemble learning with deep representation for enhanced accuracy," *Machine Learning and Knowledge Extraction*, vol. 7, no. 1, p. 4, 2025.

[25] A. S. Rodin, A. Carvalho, and A. H. Castro Neto, "Strain-induced gap modification in black phosphorus," *Physical Review Letters*, vol. 112, p. 176801, 2014.

[26] V. Tran, R. Soklaski, Y. Liang, and L. Yang, "Layer-controlled band gap and anisotropic excitons in few-layer black phosphorus," *Physical Review B*, vol. 89, p. 235319, 2014.

[27] J. Kim, S. S. Baik, S. H. Ryu, Y. Sohn, S. Park, B.-G. Park, J. Denlinger, Y. Yi, H. J. Choi, and K. S. Kim, "Observation of tunable band gap and anisotropic Dirac semimetal state in black phosphorus," *Science*, vol. 349, no. 6249, pp. 723–726, 2015.

[28] T. Xie and J. C. Grossman, "Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties," *Physical Review Letters*, vol. 120, no. 14, p. 145301, 2018.

[29] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller, "SchNet: A deep learning architecture for molecules and materials," *The Journal of Chemical Physics*, vol. 148, no. 24, p. 241722, 2018.

[30] C. Chen, W. Ye, Y. Zuo, C. Zheng, and S. P. Ong, "Graph networks as a universal machine learning framework for molecules and crystals," *Chemistry of Materials*, vol. 31, no. 9, pp. 3564–3572, 2019.

[31] K. Choudhary and B. DeCost, "Atomistic line graph neural network for improved materials property predictions," *npj Computational Materials*, vol. 7, p. 185, 2021.

[32] C. Chen and S. P. Ong, "A universal graph deep learning interatomic potential for the periodic table," *Nature Computational Science*, vol. 2, no. 11, pp. 718–728, 2022.

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